

1(a) Give **two** reasons why some programs are written in a low-level language.

1 .-----

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2 .-----

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[2]

(b) Describe the benefits of using a compiler instead of an interpreter when writing a program.

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[3]

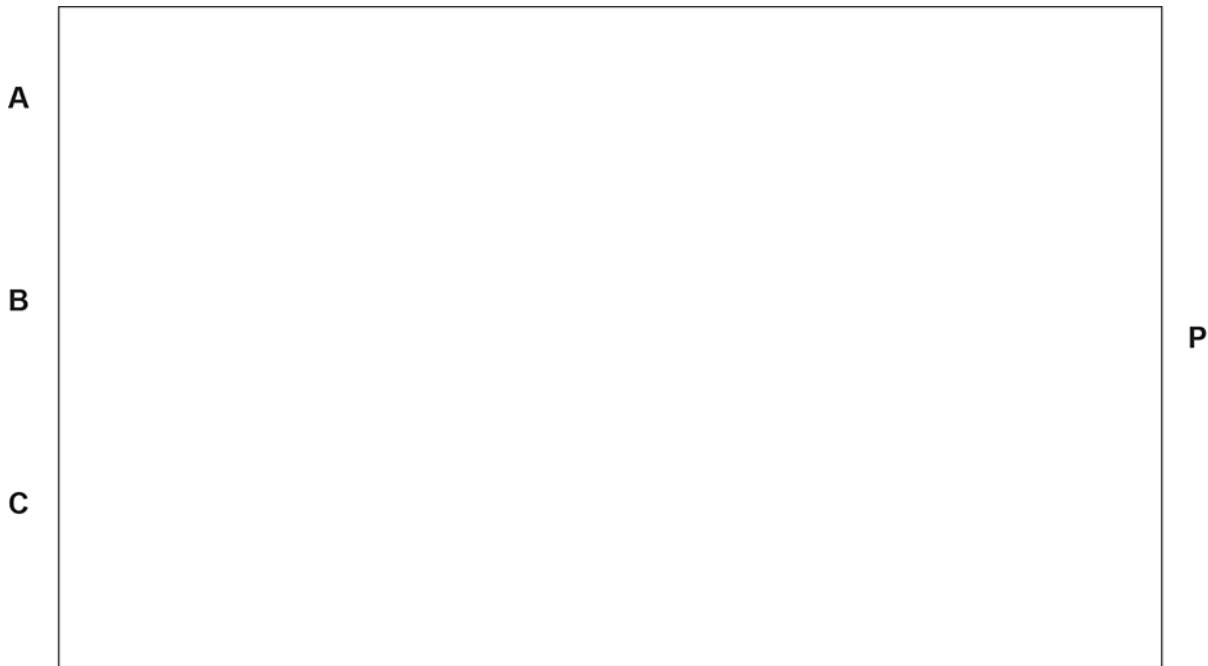
2(a) A security system protects online credit card payments. The system has three inputs:

| Input | Description                                      |
|-------|--------------------------------------------------|
| A     | True when the user enters their correct password |
| B     | True when the user enters their correct PIN      |
| C     | True when the card is blocked                    |

The output (P) is True if both of the following conditions are True:

- the user has entered either their correct password or has entered their correct PIN
- the card is **not** blocked.

Draw the logic circuit for the security system



[3]

(b) Complete the truth table for  $P = A \text{ AND } B$

| A | B | P |
|---|---|---|
|   |   |   |
|   |   |   |
|   |   |   |
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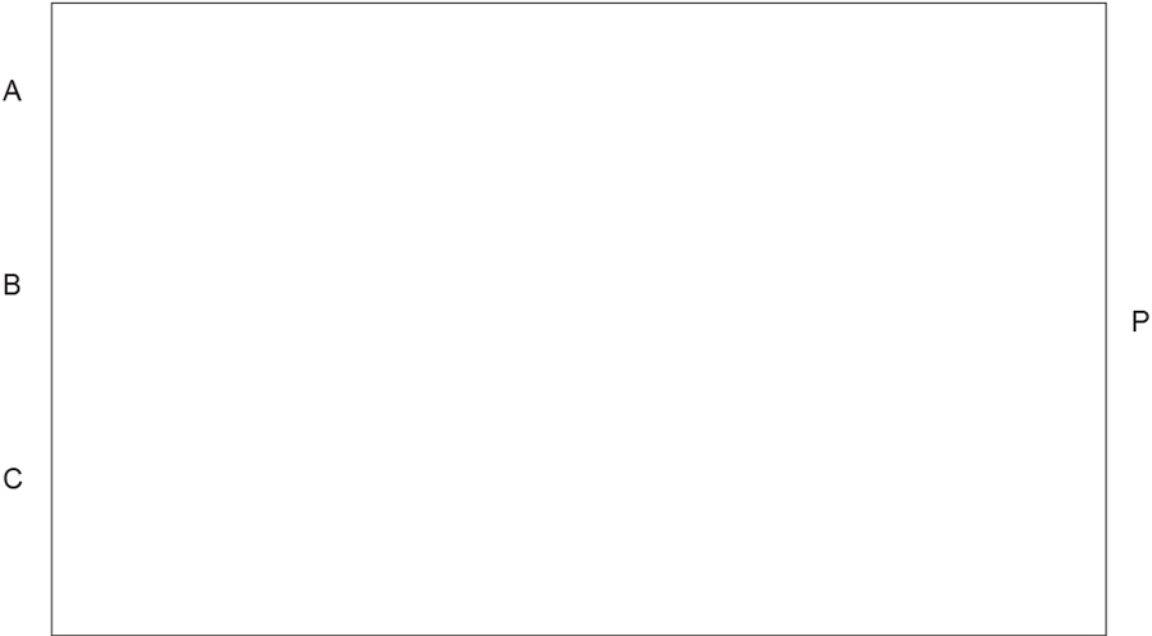
[2]

3(a) Complete the truth table for  $P = (A \text{ AND } B) \text{ OR } C$

| A | B | C | P |
|---|---|---|---|
| 0 | 0 | 0 |   |
| 0 | 0 | 1 |   |
| 0 | 1 | 0 |   |
| 0 | 1 | 1 |   |
| 1 | 0 | 0 |   |
| 1 | 0 | 1 |   |
| 1 | 1 | 0 |   |
| 1 | 1 | 1 |   |

[4]

(b) Draw a logic circuit for  $P = \text{NOT } A \text{ AND } (B \text{ OR } C)$



[3]

4 A fast food restaurant offers half-price meals if the customer is a student or has a discount card. The offer is not valid on Saturdays.

A computer system is used to identify whether the customer can have a half-price meal.

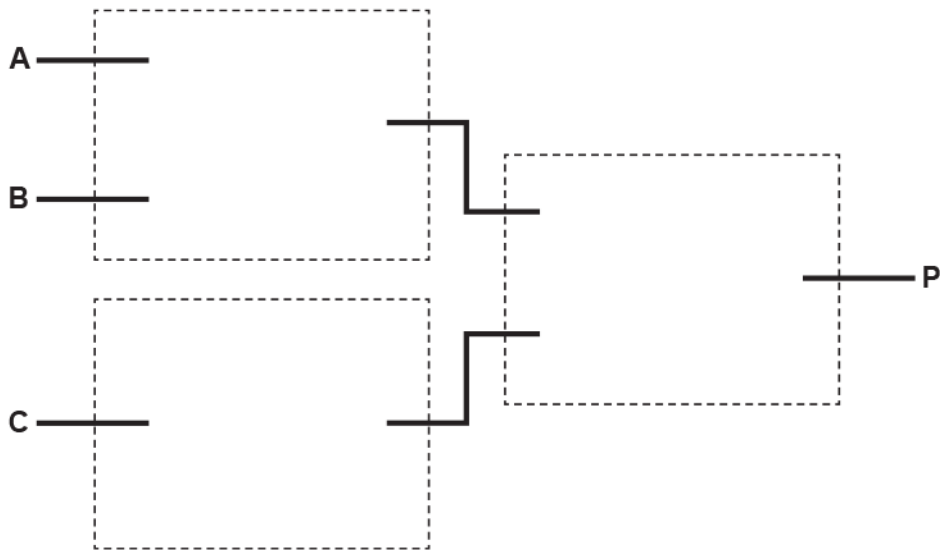
The table identifies the three inputs to the computer system:

| Input | Value               |
|-------|---------------------|
| A     | Is a student        |
| B     | Has a discount card |

|   |                             |
|---|-----------------------------|
| C | The current day is Saturday |
|---|-----------------------------|

The logic system  $P = (A \text{ OR } B) \text{ AND NOT } C$  is used.

- i. Complete the following logic diagram for  $P = (A \text{ OR } B) \text{ AND NOT } C$  by drawing one logic gate in each box.



[3]

- ii. A truth table can be produced for this logic circuit.

Describe the purpose of a truth table.

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[2]

- iii. State how many rows (excluding any headings) would be required in a truth table for the logic expression:

$$P = (A \text{ OR } B) \text{ AND NOT } C$$

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[1]

5 Identify the logic gates for truth table 1 and truth table 2.

Truth table 1:

| A | B | Output |
|---|---|--------|
| 0 | 0 | 0      |
| 0 | 1 | 1      |
| 1 | 0 | 1      |
| 1 | 1 | 1      |

Logic gate 1: .....

Truth table 2:

| A | B | Output |
|---|---|--------|
| 0 | 0 | 0      |
| 0 | 1 | 0      |
| 1 | 0 | 0      |
| 1 | 1 | 1      |

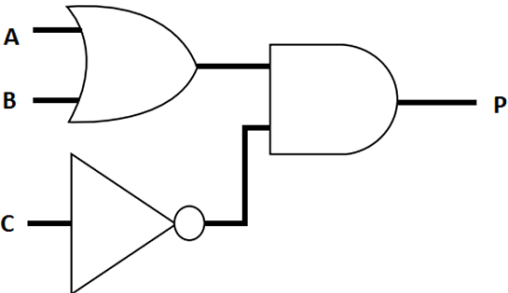
Logic gate 2: .....

[2]

END OF QUESTION PAPER

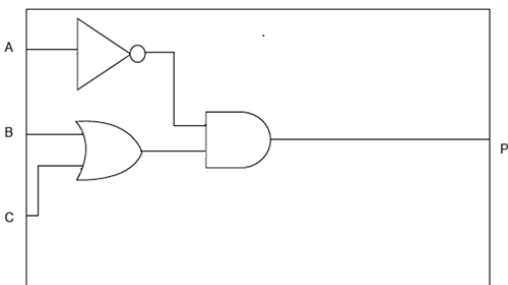
| Question |   | Answer/Indicative content                                                                                                                                                                                                                                                                                                                                                                                              | Marks              | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | a | <p>1 mark each to max 2</p> <ul style="list-style-type: none"> <li>• (machine code) does not need to be translated / compiled / interpreted</li> <li>• Direct control of <b>hardware / memory</b></li> <li>• Faster execution time</li> <li>• Code can be optimised / shorter code / use less memory</li> <li>• Can program for specific hardware</li> <li>• <b>Assembly language</b> is fast to translate.</li> </ul> | <p>2<br/>(AO1)</p> | <p>"More efficient" by itself is TV.</p> <p>Mark first answer on each line.</p> <p>BP6 relates to Assembly language being a one-to-one direct mapping to machine code.</p> <p><b><u>Examiner's Comments</u></b></p> <p>These questions assessed understanding of low-level languages and the use of compilers to translate from high-level code. Many candidates were able to do this well and clearly understood the need for high-level languages to be translated into machine code before the processor can execute this. The questions, however, did not ask just for this but specifically asked for reasons and benefits. Where candidates left their response as simply a discussion of the difference between high-level and low-level code or between translators and compilers, it was difficult for examiners to award marks.</p> <p>Better responses were able to give clear reasons for the use of low-level code (such as direct control of hardware and faster execution of code) and the benefits of using a compiler instead of an interpreter (such as being able to distribute an executable file with no access to source code and not needing to translate code again once this has been done).</p> |

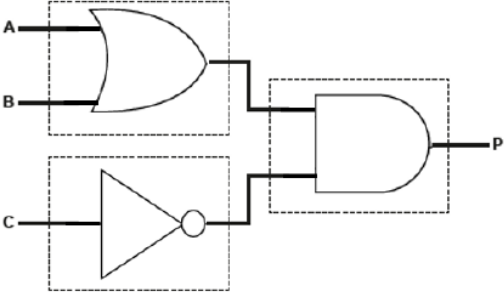
| Question |   | Answer/Indicative content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks      | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | b | <p>1 mark each to max 3</p> <ul style="list-style-type: none"> <li>• Can produce an <b>executable file</b></li> <li>• program/code <b>runs/executes</b> faster (than interpreted version)</li> <li>• end users do not need translator</li> <li>• Can be run <b>again/multiple times</b> without re-translating // only needs to translate <b>once</b></li> <li>• <b>End users</b> have no access to <b>source code</b> // distributed <b>with no source code</b>...</li> <li>• ...cannot steal/copy/modify code/program</li> <li>• Shows <b>all/multiple</b> errors // shows errors <b>at the end</b> (of compilation) // creates <b>error file</b></li> <li>• Compiler can optimise the code</li> </ul> | 3<br>(AO1) | <p>Allow in reverse (e.g. "interpreter translates every time")</p> <p>Do not allow "no access to source code" unless clearly talking about end user. Allow if in context of distribution.</p> <p>Do not allow descriptions of how a compiler translates (e.g. "translates whole code in one go")</p> <p>"Faster / quicker" by itself is TV</p> <p><b><u>Examiner's Comments</u></b></p> <p>These questions assessed understanding of low-level languages and the use of compilers to translate from high-level code. Many candidates were able to do this well and clearly understood the need for high-level languages to be translated into machine code before the processor can execute this. The questions, however, did not ask just for this but specifically asked for reasons and benefits. Where candidates left their response as simply a discussion of the difference between high-level and low-level code or between translators and compilers, it was difficult for examiners to award marks.</p> <p>Better responses were able to give clear reasons for the use of low-level code (such as direct control of hardware and faster execution of code) and the benefits of using a compiler instead of an interpreter (such as being able to distribute an executable file with no access to source code and not needing to translate code again once this has been done).</p> |
|          |   | <b>Total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>5</b>   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Question |   | Answer/Indicative content                                                                                                                                                                                                    | Marks | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2        | a | <ul style="list-style-type: none"> <li>Any use of an OR gate with 2 inputs</li> <li>NOT C</li> <li>Correctly joined by AND gate</li> </ul>  | 3     | <p>Gate diagrams must be correct. NOT must include circle; AND / OR must <b>not</b> include circle.</p> <p>FT for BP3 if appropriate (e.g. missing/wrong NOT gate, wrong gate used instead of OR)</p> <p>Max 2 if extra gates or system is not logically correct.</p> <p><b><u>Examiner's Comments</u></b></p> <p>This question was extremely well answered by candidates, showing excellent understanding of constructing logic gate diagrams from a real-world scenario.</p> <p>Misconceptions or mistakes were rare, but where they did occur they tended to be around shapes of gates. Candidates should be aware that the circle on the NOT gate is crucial (as otherwise a triangle with no circle is simply a buffer) and that AND and OR gates do not need circles, or they would instead of NAND or NOR gates.</p> <p>Examiners are instructed to mark the shapes of the gates and ignore any annotation, so it is crucial that candidates can accurately draw shapes that are clearly AND and OR gates.</p> |



| Question |   |   | Answer/Indicative content                                                                                                                                                                                             | Marks | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|----------|---|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
|          | b |   | <ul style="list-style-type: none"><li>• All input possibilities for A, B (00, 01, 10, 11), does not have to be in order...</li><li>• ...Correct output for P (only 1 when A and B are both 1, 0 otherwise).</li></ul> | 2     | <p>P = A AND B</p> <table><tr><th>A</th><th>B</th><th>P</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table> <p>Accept T/F, Yes/No, ticks/crosses, etc.</p> <p><u>Examiner's Comments</u></p> <p>This question was also extremely well answered. Most candidates filled in columns A and B with every possible combination of inputs. 1s and 0s were the expected options, but Yes/No and True/False were also accepted if given by candidates.</p> <p>Inputs do not have to be in any particular order, but best practice to start with 00 and then essentially count up in binary, through 01, 10 then 11. This means that where larger systems are presented (perhaps with 3 or 4 inputs), no input combination is missed.</p> <p>The final column (P) should have contained the output for the system, and with P = A AND B, this means that only the row with A and B as both 1 should have had a 1 output, with every other output being 0.</p> | A | B | P | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 |
| A        | B | P |                                                                                                                                                                                                                       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0        | 0 | 0 |                                                                                                                                                                                                                       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0        | 1 | 0 |                                                                                                                                                                                                                       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1        | 0 | 0 |                                                                                                                                                                                                                       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1        | 1 | 1 |                                                                                                                                                                                                                       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|          |   |   | Total                                                                                                                                                                                                                 | 5     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

| Question |   | Answer/Indicative content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks      | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |                          |
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| 3        | a | <p>1 mark per group of 2 rows</p> <table><tr><th>A</th><th>B</th><th>C</th><th>P</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table> <p>1 mark<br/>1 mark<br/>1 mark<br/>1 mark</p> | A          | B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | C | P | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 4<br>(AO2) | Accept True / False etc. |
| A        | B | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | P          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |                          |
| 0        | 0 | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |                          |
| 0        | 0 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |                          |
| 0        | 1 | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |                          |
| 0        | 1 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |                          |
| 1        | 0 | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |                          |
| 1        | 0 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |                          |
| 1        | 1 | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |                          |
| 1        | 1 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |                          |
|          | b | <p>1 mark each</p> <ul style="list-style-type: none"><li>• NOT A</li><li>• B OR C</li><li>• AND gate with <b>two</b> inputs</li></ul>                                                                                                                                                                                                                                                                                                                                   | 3<br>(AO3) | <p><b>Max 2</b> if not logically correct or any additional / missing gates.</p> <p>Shapes of gates must be correct with correct number of inputs. Ignore annotation of gate names.</p> <p>NOT gate must include circle. Other gates must not include circle.</p> <p><b>Examiner's Comments</b></p> <p>Drawing logic circuits is now a commonly asked question and candidates are generally competent at doing this. Where issues did arise, they tended to be missing the circle from the NOT gate (and therefore turning it into a buffer, not a logic gate) or including the incorrect number of inputs into a gate.</p> <p><b>Key point – logic gates</b></p> <p>Candidates are expected to be able to draw the correct shapes for each gate. A number of candidates labelled their gates up, but this is not necessary; examiners are instructed to mark the shape of each gate and ignore any labelling.</p> <p>Candidates are <u>not</u> allowed to take stencils or other tools that allow them to more easily draw these gates into an examination unless as part of a specific access arrangement agreed by OCR.</p> |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |                          |
|          |   | Total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 7          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |                          |

| Question | Answer/Indicative content                                                                                                                                                                  | Marks            | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4        | <p>i</p> <ul style="list-style-type: none"> <li>• A OR B</li> <li>• NOT C</li> <li>• AND gate</li> </ul>  | 3<br>(AO2<br>1b) | <p>1 mark per gate. Correct symbols must be used.</p> <p>NOT gate must have circle for inversion, OR and AND must not have a circle</p> <p>Mark the <b>shape</b> of each gate, not the name written if given. Ignore any writing / notes.</p> <p>Lines do not have to be drawn or joined up, but if they are, gates must have the <b>correct number of inputs/outputs</b>. Penalise once then FT.</p> <p><b>Examiner's Comments</b></p> <p>This question was answered extremely well. Candidates were familiar with the logic symbols required for each gate.</p> <p>A few candidates missed the circle from the NOT gate symbol (which turned this into a buffer instead) or added a circle to the AND and OR gates. This turned them into NAND and NOR gates. These logic gates are not on the GCSE specification, but are covered at A Level. Teachers may find it beneficial to briefly discuss these in order to anticipate mistakes like these.</p> <p>A small number of candidates drew indistinct logic gate symbols and named them instead, such as a rectangle with AND written in; this was not credited with a mark.</p> |

| Question |  |     | Answer/Indicative content                                                                                                                                                              | Marks         | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|--|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |  | ii  | <ul style="list-style-type: none"> <li>To show <b>all possible inputs</b> (to the logic circuit)...</li> <li>...and the <b>associated/dependent</b> output (for each input)</li> </ul> | 2<br>(AO1 1b) | <p>For 2<sup>nd</sup> BP, must be clear that the output is linked to the input values given.</p> <p>“All possible combinations of inputs and outputs” gains the first mark (all possible inputs) but not the second.</p> <p>“The output for each possible input” gains both marks</p> <p><b><u>Examiner’s Comments</u></b></p> <p>It is clear that candidates have experienced and used truth tables with logic circuits. However, the majority were not able to describe the truth table’s purpose with precision.</p> <p>A truth table defines the expected outputs for a logic circuit depending on the inputs given. Furthermore, the truth table covers all possible permutations of inputs.</p> <p>A logic circuit with three inputs (A,B and C) will have <b>8</b> possible rows in a truth table to cover the 8 possible ways that True and False values for three inputs can be arranged.</p> <div style="text-align: center;">  <b>Misconception</b> </div> <p>The number of rows in a truth table depends on the number of inputs. The number of rows can be given by <math>2^x</math>, where x is the number of inputs. Therefore, a truth table with 3 inputs would have <math>2^3</math> rows, or <math>2 \times 2 \times 2 = 8</math> rows.</p> <p><b>Exemplar 1</b></p> <p><i>To show what output happens for every variation of inputs and check they are all what was intended</i></p> <p>Here the candidate has achieved both marks available; it is clear that every variation of input is included in the truth table and the output is very clearly linked to these inputs.</p> |
|          |  | iii | <ul style="list-style-type: none"> <li>8 // eight</li> </ul>                                                                                                                           | 1<br>(AO2 1a) | <p>Accept other answers that equate to 8 (e.g. <math>2^3</math>)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Question |  |  | Answer/Indicative content | Marks | Guidance |
|----------|--|--|---------------------------|-------|----------|
|          |  |  | Total                     | 6     |          |

| Question |  |  | Answer/Indicative content                                                                                               | Marks         | Guidance                                                                                                                                                                                                                                                                                                                                               |
|----------|--|--|-------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5        |  |  | 1 mark each <ul style="list-style-type: none"> <li>Logic gate 1: <b>OR</b></li> <li>Logic gate 2: <b>AND</b></li> </ul> | 2<br>(AO2 1a) | Allow <b>A OR B</b> // <b>B OR A</b> for logic gate 1<br>Allow <b>A AND B</b> // <b>B AND A</b> for logic gate 2<br><br>If logic statement provided with multiple gates (e.g. <b>A OR B AND C</b> ) this is incorrect.<br><br>Allow use of symbols (e.g. $\vee$ , $+$ for OR, $\wedge$ , $\cdot$ for AND)<br><br>Allow correct drawing of logic gates. |
|          |  |  | <b>Total</b>                                                                                                            | <b>2</b>      |                                                                                                                                                                                                                                                                                                                                                        |